Gradient and Jacobian

math-foundations / concept 14

First, an example

Consider the bowl-shaped scalar field $f(x, y) = x^2 + y^2$. Its partial derivatives are $\partial f/\partial x = 2x$ and $\partial f/\partial y = 2y$, so the gradient is the vector field

$$\nabla f(x, y) = (2x,\ 2y).$$

Read aloud: "nabla f at x, y equals the vector two-x, two-y."

At the point $(3, 4)$ this gives $\nabla f(3, 4) = (6, 8)$, a vector of length $\sqrt{6^2 + 8^2} = 10$ pointing radially outward from the origin. This is the direction of steepest ascent: moving an infinitesimal step along $(6, 8)$ increases f faster than any other direction of equal length, and the rate of that increase is 10. Negating it gives the direction of steepest descent — the move that gradient descent would take from $(3, 4)$. The rest of this lesson develops the gradient and its vector-valued cousin, the Jacobian, in full generality.

1 Setup and motivation

We work in finite-dimensional Euclidean space $\mathbb{R}^n$ with the standard inner product $\langle u, v\rangle = u^\top v$ and induced norm $\|v\| = \sqrt{\langle v, v\rangle}$. For scalar functions, the derivative captures a single rate of change; for vector-valued functions, it captures *all* pairwise rates between input and output coordinates, organized into a matrix.

2 Definitions

Definition 1 (Partial derivative)**.** Let $U \subseteq \mathbb{R}^n$ be open and $f : U \to \mathbb{R}$. The *partial derivative* of f with respect to x_j at $x \in U$ is

$$\frac{\partial f}{\partial x_j}(x) = \lim_{h\to 0} \frac{f(x + he_j) - f(x)}{h},$$

Read aloud: "partial f by partial x-j at x equals the limit, as h tends to zero, of f of x plus h times e-j minus f of x, all divided by h."

where e_j is the jth standard basis vector, when this limit exists.

Definition 2 (Gradient)**.** Let $f : U \subseteq \mathbb{R}^n \to \mathbb{R}$ be differentiable at x. The *gradient* of f at x is the column vector

$$\nabla f(x) \ = \ \left(\tfrac{\partial f}{\partial x_1}(x),\ \dots,\ \tfrac{\partial f}{\partial x_n}(x)\right)^\top \in \mathbb{R}^n.$$

Read aloud: "nabla f at x equals the column vector of partial f by partial x-1 through partial f by partial x-n, transposed, an element of R-to-the-n."

Equivalently, $\nabla f(x)$ is the unique vector such that $D_v f(x) = \langle \nabla f(x), v\rangle$ for every $v \in \mathbb{R}^n$, where $D_v f(x) = \lim_{t\to 0}\big(f(x + tv) - f(x)\big)/t$ is the directional derivative.

Definition 3 (Jacobian matrix)**.** Let $F : U \subseteq \mathbb{R}^n \to \mathbb{R}^m$ with components $F = (F_1, \dots, F_m)$, differentiable at x. The *Jacobian matrix* of F at x is the $m \times n$ matrix

$$J_F(x) \; = \begin{pmatrix} \dfrac{\partial F_1}{\partial x_1}(x) & \cdots & \dfrac{\partial F_1}{\partial x_n}(x) \\ \vdots & \ddots & \vdots \\ \dfrac{\partial F_m}{\partial x_1}(x) & \cdots & \dfrac{\partial F_m}{\partial x_n}(x) \end{pmatrix}, \qquad [J_F(x)]_{ij} = \frac{\partial F_i}{\partial x_j}(x).$$

Read aloud: "the i-j entry of the Jacobian J-F at x is partial F-i by partial x-j."

$J_F(x)$ is the matrix of the total derivative $dF_x : \mathbb{R}^n \to \mathbb{R}^m$ in standard bases.

Definition 4 (Jacobian determinant)**.** If $m = n$, the *Jacobian determinant* of F at x is $\det J_F(x) \in \mathbb{R}$. It is the signed factor by which F locally scales n-dimensional volume at x: a small parallelepiped of volume V at x has image of volume $|\det J_F(x)|\, V + o(V)$.

3 The gradient and steepest ascent

Lemma 1 (Directional-derivative formula)**.** *If f is differentiable at x, then for every $v \in \mathbb{R}^n$,*

$$D_v f(x) = \langle \nabla f(x), v \rangle.$$

Read aloud: *"the directional derivative D-v of f at x equals the inner product of nabla f at x with v."*

Sketch. Differentiability at x means $f(x+h) = f(x) + \langle \nabla f(x), h \rangle + r(h)$ with $r(h)/\|h\| \to 0$. Setting $h = tv$ and dividing by t gives the claim in the limit $t \to 0$. □

Theorem 2 (Direction of maximum directional derivative)**.** *Let $f : U \subseteq \mathbb{R}^n \to \mathbb{R}$ be differentiable at $x \in U$, with $\nabla f(x) \neq 0$. Then among all unit vectors $v \in \mathbb{R}^n$ (i.e. $\|v\| = 1$), the directional derivative $D_v f(x)$ attains its maximum at*

$$v^\star = \frac{\nabla f(x)}{\|\nabla f(x)\|},$$

Read aloud: *"v-star equals nabla f at x divided by the norm of nabla f at x."*

and the maximum value is $\|\nabla f(x)\|$.

Proof. By Lemma 1, $D_v f(x) = \langle \nabla f(x), v \rangle$. The Cauchy–Schwarz inequality states that for any $u, v \in \mathbb{R}^n$,

$$|\langle u, v \rangle| \;\le\; \|u\|\,\|v\|,$$

Read aloud: "the absolute value of the inner product of u and v is at most the norm of u times the norm of v."

with equality iff u and v are linearly dependent. Applying this with $u = \nabla f(x)$ and a unit vector v,

$$D_v f(x) = \langle \nabla f(x), v \rangle \;\le\; \|\nabla f(x)\|\,\|v\| = \|\nabla f(x)\|.$$

Read aloud: "D-v f at x equals the inner product of nabla f at x with v, which is at most the norm of nabla f at x times the norm of v, equal to the norm of nabla f at x."

Equality holds iff v is a positive scalar multiple of $\nabla f(x)$; since $\|v\| = 1$ and $\nabla f(x) \neq 0$, the unique maximizer is $v^\star = \nabla f(x)/\|\nabla f(x)\|$. Substituting back gives $D_{v^\star} f(x) = \langle \nabla f(x), \nabla f(x)/\|\nabla f(x)\| \rangle = \|\nabla f(x)\|$. □

Example 1. For $f(x, y) = x^2 + 3y^2$, $\nabla f(x, y) = (2x, 6y)$. At $(1, 1)$, the steepest-ascent direction is $(2, 6)/\sqrt{40} = (1, 3)/\sqrt{10}$ with rate $\sqrt{40}$, while $-\nabla f$ points toward the minimum at the origin.

4 The chain rule for vector-valued maps

Theorem 3 (Chain rule)**.** *Let $G : \mathbb{R}^p \to \mathbb{R}^n$ be differentiable at x, and $F : \mathbb{R}^n \to \mathbb{R}^m$ differentiable at $y = G(x)$. Then $H = F \circ G$ is differentiable at x and*

$$J_H(x) \;=\; J_F(G(x)) \cdot J_G(x),$$

Read aloud: *"the Jacobian of H at x equals the Jacobian of F at G of x, times the Jacobian of G at x."*

an $m \times p$ matrix obtained as the product of an $m \times n$ matrix and an $n \times p$ matrix.

This is the algebraic backbone of reverse-mode automatic differentiation: for a deep composition $F_L \circ F_{L-1} \circ \cdots \circ F_1$, the overall Jacobian factors as a product of layer Jacobians, and the loss gradient propagates by successive vector–Jacobian products.

5 Worked example: $F(x, y) = (x^2 - y, e^{xy})$

The partials are $\partial F_1/\partial x = 2x$, $\partial F_1/\partial y = -1$, $\partial F_2/\partial x = ye^{xy}$, $\partial F_2/\partial y = xe^{xy}$. Hence

$$J_F(x,y) = \begin{pmatrix} 2x & -1 \\ ye^{xy} & xe^{xy} \end{pmatrix}, \qquad J_F(1,0) = \begin{pmatrix} 2 & -1 \\ 0 & 1 \end{pmatrix}, \quad \det J_F(1,0) = 2.$$

Read aloud: "J-F at x, y is the two-by-two matrix with rows two-x, minus one and y-e-to-the-x-y, x-e-to-the-x-y; at one-zero this becomes two, minus one, zero, one, with determinant two."

By the inverse function theorem, F is a local diffeomorphism near $(1, 0)$, locally doubling area.